

Reconsidering Pink-Collar Work: Evidence from Chinese Trade Shocks on Gender Occupational Segregation

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Abstract

Building on an influential body of work exploring the effects of Chinese import penetration on US manufacturing, this thesis exploits the gender-specific components of trade shocks to examine whether and how import competition reshapes local labor market occupational segregation. Using US Census and UN Comtrade data from 1990-2014, I measure commuting-zone variation in Chinese import exposure instrumented by trade flows to other high-income countries to explore shifts in occupational segregation. I find male-specific trade shocks reduce occupational segregation, while female-specific trade shocks increase occupational segregation by comparable magnitudes. Among female-dominated jobs, male-specific trade shocks are associated with a nearly 29 percentage point drop in occupational segregation. However, increased male-specific trade penetration is associated with a 3.5 percentage point drop in the share of men in pink-collar jobs. These results indicate that trade-driven declines in occupational segregation operate through occupation mix adjustments rather than male entry into female-dominated work. This distinction has important implications for understanding how trade shocks shape gender inequality in US local labor markets.

1 Introduction

Despite substantial progress in women’s social and economic standing in recent decades, significant gender gaps in economic equality remain. Among the most persistent drivers of gender economic inequality is occupational segregation; according to Blau and Kahn [2017], the share of the gender wage gap explained by occupational differences increased from 27 percent in 1980 to 49 percent in 2010. Several reasons have been proposed to explain why men and women choose different jobs—innate preferences, differences in risk-taking, competitiveness, beliefs about returns to skill, etc.—but the issue of segregation remains a significant contributor to gender inequality in high-income countries today [OECD, 2019].

At the same time, increased competition in global markets may be changing the nature of work. In the past decade, a significant body of research has emerged discussing the potential gender effects of Chinese trade liberalization [Autor et al., 2017, Cortes and Pan, 2019, Black and Brainerd, 2002, Delfino, 2020]. Since the 1980’s, mid-skill jobs have been declining as rapid polarization of labor market growth biases toward high- and low-skill work, which has been especially detrimental to manufacturing jobs [Autor et al., 2013]. This shift has been caused in part by growing trade competition from China. Higher international competition allows firms to cut costs and outsource certain mid- and low-skilled jobs to foreign competition, resulting in a decline in industries such as manufacturing [Autor et al., 2017]. Manufacturing is a predominantly male industry, therefore a loss of jobs in manufacturing primarily affects male workers. At the same time, female-dominated sectors of service and clerical work are in a boom. With the rise of female-dominated jobs (often called “pink-collar” work) and the fall of male-dominated fields such as manufacturing, one may expect existing occupational segregation to fall as unemployed male workers flock to rapidly-growing pink-collar jobs.

Occupational segregation decline plateaued after the 1990s, after seeing its largest decline during the 1970s, which was primarily driven by women entering male-dominated spaces such as law, engineering, and business [Cotter et al., 1995, Blau et al., 2013]. In recent years, more literature has emerged discussing the opposite—male entries to female-dominated occupations [Arndt and Bigelow, 2005, Delfino, 2020, Froehlich et al., 2020]. However, as pink-collar work is often stigmatized, many scholars believe male entrants are less likely to take these low-social or low-financial opportunity jobs. Still, due to the fall of manufacturing, some men may have to choose between entering a stigmatized pink-collar job or unemployment.

Seeing these trends in pink-collar work and manufacturing decline, this thesis asks the question: does increased international competition with China affect occupational segregation, particularly in female-dominated fields? Following an approach pioneered by Autor et al. [2017], I answer this question in local labor markets using US Census data from 1990-2014 and UN Comtrade data on US-China manufacturing import pene-

tration [Autor et al., 2017]. I find that increased import penetration in manufacturing is not linked to an overall decrease in gender segregation, but is associated with a significant decline in occupational segregation when isolating male-specific trade shocks. These effects are even greater within pink-collar jobs, where trade shocks are associated with a substantial drop in occupational segregation. However, I find that import penetration is not associated with an overall increase in the share of men in pink-collar jobs, suggesting declines in occupational segregation can be attributed to a change in commuting-zone-level occupational mix rather than within-job sex composition.

The remainder of this thesis is organized as follows. Section 2 describes the theoretical and conceptual framework of pink-collar work and its relation to Chinese trade shocks. Section 3 describes my data and methodology. Section 4 discusses my empirical design for Section 5, which shows my results and a discussion thereof. I conclude in Section 6 and discuss further possible research implications of my results.

2 Conceptual Framework

Gender occupational segregation is an important measure of gender equality in an economy, as it has numerous implications on women’s labor force participation and wages [Blau et al., 2013]. Significant scholarship has examined the implications of women entering male-dominated fields (such as law, engineering, and business) during the 1980s and 1990s, though less attention has been paid to the process of men entering female-dominated occupations [Bianchi and Rytina, 1986, Cotter et al., 1995, Blau et al., 2013]. However, as historically-female sectors such as service and clerical work continuously grow and historically-male sectors such as manufacturing suffer massive decline, the future of pink-collar work will have significant implications for gender equality [Duran-Franch, 2020]. This thesis focuses on female-dominated, or “pink-collar,” work. In doing so, I define and discuss the nature of pink-collar work before examining Chinese trade shocks and their potential effects on occupational segregation.

2.1 Defining Pink-Collar Work

Even as gender occupational segregation has decreased in past decades, a significant share of the workforce remains divided by gender. For most of the twentieth century, women’s labor market participation was constrained by social norms and childrearing responsibilities, particularly at marriage, where the (positive) substitution effects of women’s wages were dwarfed by the (negative) income effects of their husband’s earnings [Goldin, 2006]. As women entered the workforce in unprecedented numbers in the second half of the twentieth century, these limitations constrained women who worked to stay in jobs that were seen as “respectable”—professional or clerical work with short or flexible hours, such as primary school teaching or

secretary work [Goldin, 2006]. The widespread diffusion of the birth control pill after the 1970s solved a key timing problem, as women could now delay childbearing while expecting their peers would do the same, easing the stigma against staying in school or in the labor market during their early 20s [Goldin and Katz, 2000]. This increased labor-market attachment opened the door to career paths previously accessible only to men.

In literature examining pink-collar work, scholars usually isolate specific occupations, such as social work or primary school educators [Delfino, 2020, Lordan and Pischke, 2016]. For the purposes of this thesis, I have defined pink-collar work as any occupation that, according to a 1980 Census sample, is more than 70% female, whereas male-dominated fields are likewise defined by a 70%-male cutoff. A complete list of these occupations, including their gender share in 1980, is located in Table 9. This cut-off includes occupations traditionally considered in the literature as “pink-collar.”

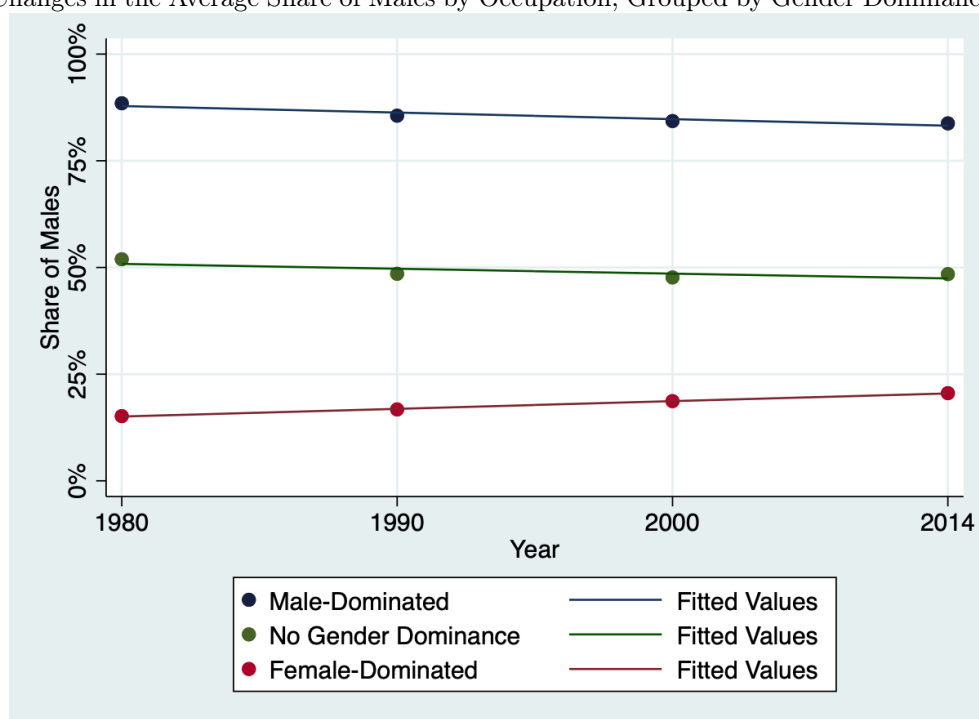
In the past twenty years, several scholars have studied the impact of social identity on occupational choice [Akerlof and Kranton, 2000, Arndt and Bigelow, 2005, Croson and Gneezy, 2009, Delfino, 2020, Eagly and Wood, 2012]. Occupation is a large part of one’s identity, as it captures a significant percentage of one’s waking hours. Social pressures to conform to masculinity and femininity can affect labor market preferences [Akerlof and Kranton, 2000]. According to gender theorists, the act of “performing gender,” or making choices aligned with certain gender roles, is an essential component of being a man or woman [Butler, 1988]. As such, an economically and socially vital decision such as occupational choice is impacted by one’s desire to conform to standards of masculinity and femininity. As Akerlof and Kranton [2000] discuss, the desire to show consistent gender displays can affect one’s perception of an occupation (for example, if men value dominant positions, assistant-level or secretarial work may threaten their identities). Thus, as the household labor capacity of women has been historically naturalized, female-dominated jobs likewise conform to traditional standards of femininity [Becker, 1974].

Still, even as women have penetrated male-dominated fields in recent years, economists have paid less attention to men entering pink-collar work. In Figure 1, I show the changes in the average share of males across occupations by year, sorted by gender dominance.

While historically some male-dominated fields (such as bank telling) have become female-dominated, the opposite almost never occurs [Arndt and Bigelow, 2005].¹ Building on the literature of sociologists, psychologists, and gender theorists, economists have proposed several reasons men may resist entering pink-

¹It is of note that gender roles are not simply binary, but hierarchical. Therefore, while women accepting male positions is an “upgrade,” albeit contrary to their gender performance, men accepting female positions is a “downgrade” and against their gender display. The latter is, therefore, more heavily penalized.

Figure 1: Changes in the Average Share of Males by Occupation, Grouped by Gender Dominance, 1980-2014



Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. *See* Ruggles et al. [2020].

Note: Share of males in each occupation by year was computed using individual-level Census data. Averages across male- and female-dominated jobs are shown here.

collar work. First, feminized labor traditionally has been socially and financially devalued. As Cohen and Huffman [2003] find, men’s superior position in labor markets may incentivize degrading the value of female work. Since men may perceive women’s work as less valuable, and therefore less financially beneficial, they may not pursue these opportunities. Secondly, men may not consider women’s work to have high enough returns to skill [Del Carpio and Guadalupe, 2018]. Women’s jobs are often portrayed as “easier” than men’s jobs, particularly those that are social-intensive rather than brawn-intensive. As men expect these jobs not to challenge them or promote their career, they may not see pink-collar work as personally beneficial. Third, and perhaps most often discussed, male workers may face a crisis in identity after losing their job, making it even more difficult to enter pink-collar work. This is a phenomenon economists have called “retrospective-wait unemployment”—unemployed manufacturing workers are waiting to find a job that does not exist in today’s globalized world [Katz, 2010]. The option of entering pink-collar work, which seems entirely foreign, interferes with one’s occupational identity.

However, Delfino [2020] finds that men respond less to the gender composition of a workforce than to its economic incentives. Rather than prioritizing their “male identity,” male candidates worry about wage differentials and growth opportunities. On the other hand, Lordan and Pischke [2016] find that female satisfaction has a negative relationship with the share of men in an occupation. Similarly, Valet [2018] finds that women in male-dominated fields (such as law and business) are more likely to view gender wage discrimination as unjust compared to women in female-dominated fields. This may occur because women in male-dominated jobs have fewer female referents, and are thus more likely to compare themselves to their male counterparts, whereas women in pink-collar jobs are less likely to compare themselves to male coworkers, thus not noticing wage inequality [Valet, 2018].

Gender is a major driver of one’s occupational choices and satisfaction. The gender implications of pink-collar work therefore may exert significant influence on men’s decision to enter these fields.

2.2 Gendered Effects of Chinese Trade Competition and Automation

A large body of work has emerged over the past two decades examining the impact of Chinese trade liberalization on US local labor markets. After China’s first transition to a market-oriented economy under Deng Xiaoping during the 1980s and realization of these reforms in the 1990s, Chinese growth skyrocketed far past initial predictions. China’s new and rapid transition to a market economy, sudden access to formerly illegal technologies and foreign inputs, and relatively low-wage labor have given it a significant comparative advantage in the production of labor-intensive goods [Autor et al., 2013]. At the same time, barriers to trade were lifted as China entered the WTO in 2001 and the US granted China “permanent normal trade

relations.” These changes have put significant pressures on US industries, most substantially discussed across US manufacturing [Autor et al., 2016, Pierce and Schott, 2016]. As concerns over the US economy’s rising wage inequality and “employment sag” grew during the 2000s, several economists examined the effect of Chinese import competition on changes in the US economy. Between 2001 and 2007 alone, US manufacturing employment fell 18 percent [Pierce and Schott, 2016]. In June 2015, manufacturing only composed 8.6% of non-farm employment, compared to 39.0% in January of 1944 [Autor et al., 2016]. At the same time, between 1990 and 2011, China increased its share of world manufacturing exports from 2% to 16% as overall average annual growth in exports remained at a high level [Hanson, 2012].

Although manufacturing decline greatly affects male workers, as manufacturing is a predominantly-male industry, recent literature has examined the various gender effects of this decline [Autor et al., 2017]. Autor et al. [2017] discuss the impact of manufacturing import penetration from China on men’s marriage market prospects. Most notably, they find manufacturing decline decreases the number of “marriageable” men and results in higher shares of unwed and single-parent households, affirming sociologist William Julius Wilson’s marriage model [Wilson, 1997]. Whereas Autor et al. [2017] focus on the decline of manufacturing and male workers, several other scholars examine the impact of trade shocks on women’s work. Black and Brainerd [2002] find that increased trade competition may benefit women by decreasing firms’ ability to discriminate, particularly in industries that experienced a trade shock. Cortes and Pan [2019] discuss the impact of routinization on pink-collar work, particularly clerical work that can be automated, which may be influencing gender polarization of occupations. Similarly, Ngai and Petrongolo [2017] find that, whereas male job losses are occurring in manufacturing, the net increase in female hours worked is occurring in the growing service sector, where women may retain a comparative advantage. Duran-Franch [2020], similar to this thesis, studies how a decline in employment in blue-collar work may push men to work in service and clerical work, traditionally pink-collar jobs. Building on this body of work, this thesis isolates the effect of trade shocks on occupational segregation, particularly in pink-collar occupations.

3 Data Description

3.1 Data

Data on gender segregation was sourced from IPUMS, using the 5% state samples for 1980 and 1990, the 5% sample from 2000, and the 2013-15 ACS samples for 2014 [Ruggles et al., 2020]. I drop from my analysis individuals with unidentified occupations and unemployed persons.² The 1990-2014 time frame was chosen

²Individuals identified as unemployed or not in the labor force in each Census sample were dropped to observe segregation within the occupation at the time, however I include results for occupational segregation including unemployed individuals

to include pre-shock controls for gender dominance from the 1980 data, as well as analysis within 1990-2014.

Building on the approach of Autor et al. [2017], I transformed my data into commuting-zone level data using a crosswalk that covers 722 commuting zones, spanning all of the mainland United States.³ Estimating trade effects on the commuting-zone level presents several benefits. First, commuting zones allow for analysis across the United States rather than narrowing analysis to large metropolitan areas. Secondly, commuting zones give a better picture of local labor market geography by better replicating the relationship between work and place [Tolbert et al., 1996]. Finally, most relevant to this analysis, Chinese import penetration varies widely across metropolitan areas [Autor et al., 2016]. This variation is helpful for understanding the impacts of Chinese trade shocks across the US. Limits to this approach include labor market movement and changes, but research shows regional labor market adjustments are generally slow [Autor et al., 2016]. With these benefits in mind, Census samples on the state- and county-level were transformed into commuting-zone level data across 722 commuting zones spanning the contiguous United States.

I measure occupational gender segregation using Duncan Indices on the occupation-level in each commuting zone. Duncan Indices, or gender dissimilarity indices, compute segregation of a binary group [Duncan and Duncan, 1955]. While there are other indices of segregation, such the indices of exposure or isolation, the Duncan Index is best used for a binary variable of interest, such as sex, across several groups, such as occupations. Further, most relevant literature on occupational segregation uses the Duncan Index as its measure of dissimilarity [Cortes and Pan, 2019, Bianchi and Rytina, 1986, Cotter et al., 1995, Blau et al., 2013]. The formula for the Duncan Index is as follows:

$$D = \frac{1}{2} \sum_{i=1}^N \left| \frac{m_i}{M} - \frac{f_i}{F} \right| \times 100$$

Where m_i (f_i) represents the number of male workers in the subgroup i and M (F) represents the total number of males (females) in the group. The Duncan Index varies between 0 and 100, where 100 represents perfect inequality and 0 perfect equality.⁴ In theory, if $D = \delta$, then $\delta\%$ of workers in the economy would have to switch jobs in order to achieve perfect integration. Changes in overall market-wide occupation-level Duncan Indices over time are plotted in Figure 2.

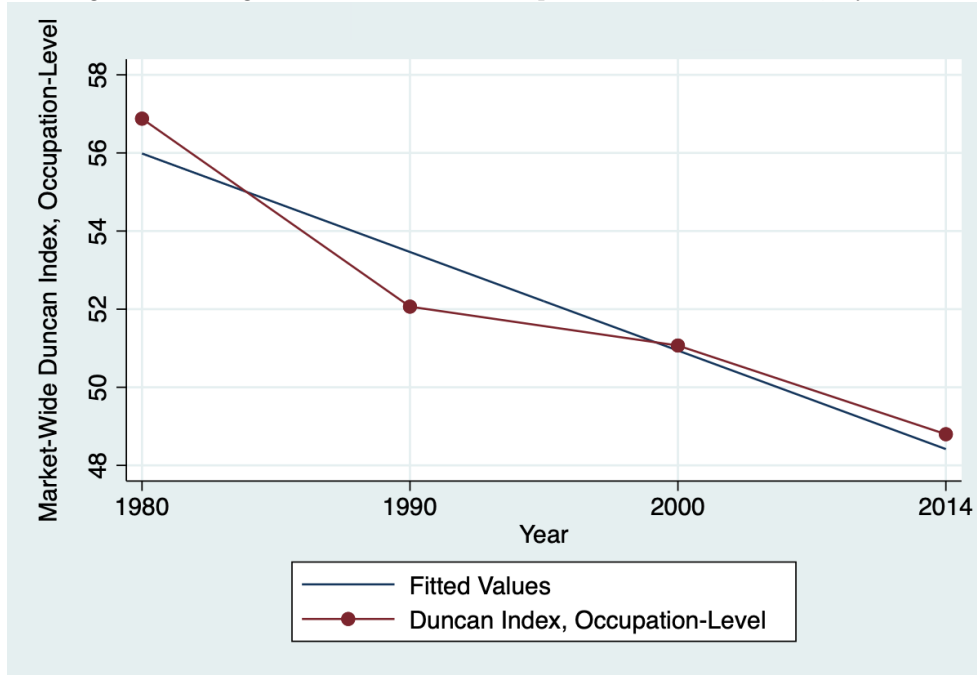
I choose to examine occupational gender segregation over other measures, such as industry-level segregation, as granularity in subgroups can better display the real degree of segregation by gender. I show this

and individuals not in the labor force in Appendix Tables 10 through 15. These results may capture an “identity shift” in occupational segregation, since this would include people who identify with a certain occupation, regardless of whether or not they are working at the time. Still, I drop unemployed persons from observation to measure the reality of segregation in these occupations. Notably, dropping these observations does not change my conclusions throughout this paper, which can be seen in the results shown in the Appendix.

³Alaska and Hawaii are not included in my analysis, so clustered errors by state include 48 clusters.

⁴I multiply the Duncan Index by 100 for ease of interpretation. Traditionally, the Duncan Index ranges between 0 and 1.

Figure 2: Changes in Market-Wide Occupational Duncan Indices, By Year



Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020].

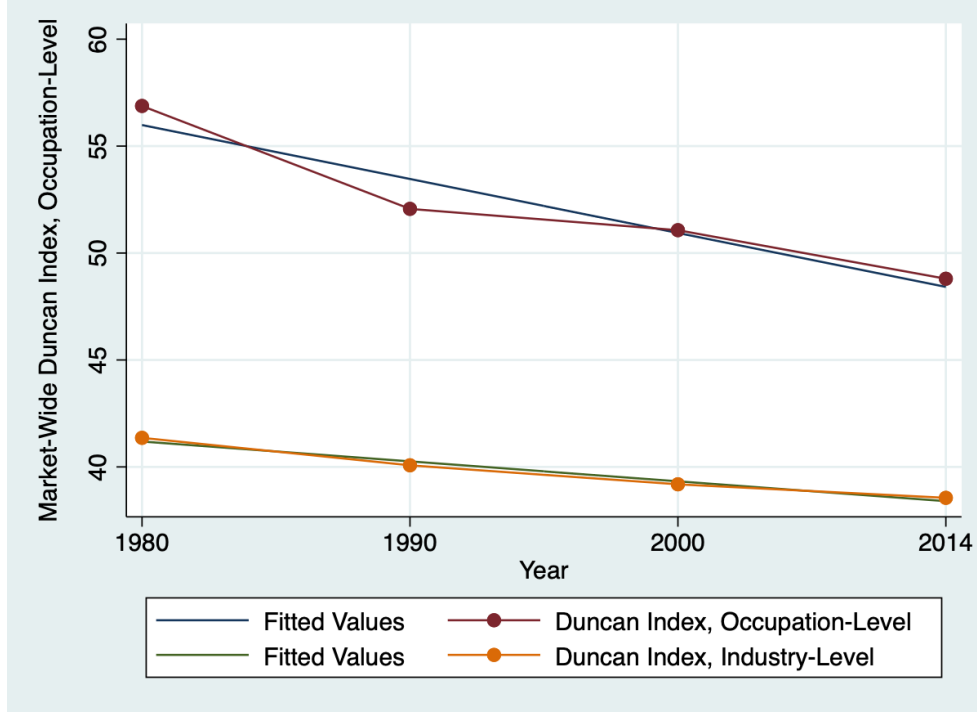
Note: Occupational Duncan Indices are shown above, charted over Census sample year. This analysis leaves out occupational codes for unknown/unclassified occupations and unemployed persons.

phenomenon in Figure 3. While taken on the same samples, since there are fewer subgroups within industries compared to occupations, industry-level segregation is smaller than occupational segregation over the same period. In order to better represent the actual degree of segregation in commuting zones without distorting samples by choosing so small a subgroup, I examine occupational segregation over industry-level segregation or other subgroups. Further, pink-collar designations are given to occupations rather than industries. For example, in 1980, Industry “Metalwork Machinery” contained both female- and male-dominated occupations such as secretaries and production supervisors. Examining this industry would therefore obscure actual differences in gender segregation.

In addition to Duncan Indices, I also constructed several measures of the gender mix in each occupation. To examine changes in occupational mix in predominantly female industries, I used the average share of females or males in an occupation, according to the 1980 Census sample, to label these female-dominated or male-dominated jobs. I have provided a table with these occupations in Table 5.

There are significantly fewer female-dominated jobs than male-dominated jobs, as shown in Figure 4. Across the 382 included occupational codes, only around 17% of these are 70% or more female. Still, these occupations are not necessarily “rarer” than male-dominated jobs. As I show in Figure 5, in the 1980 Census sample, while female-dominated jobs still account for fewer occupational codes than male-dominated jobs,

Figure 3: Changes in Segregation across Occupations and Industries, 1980-2014



Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020].

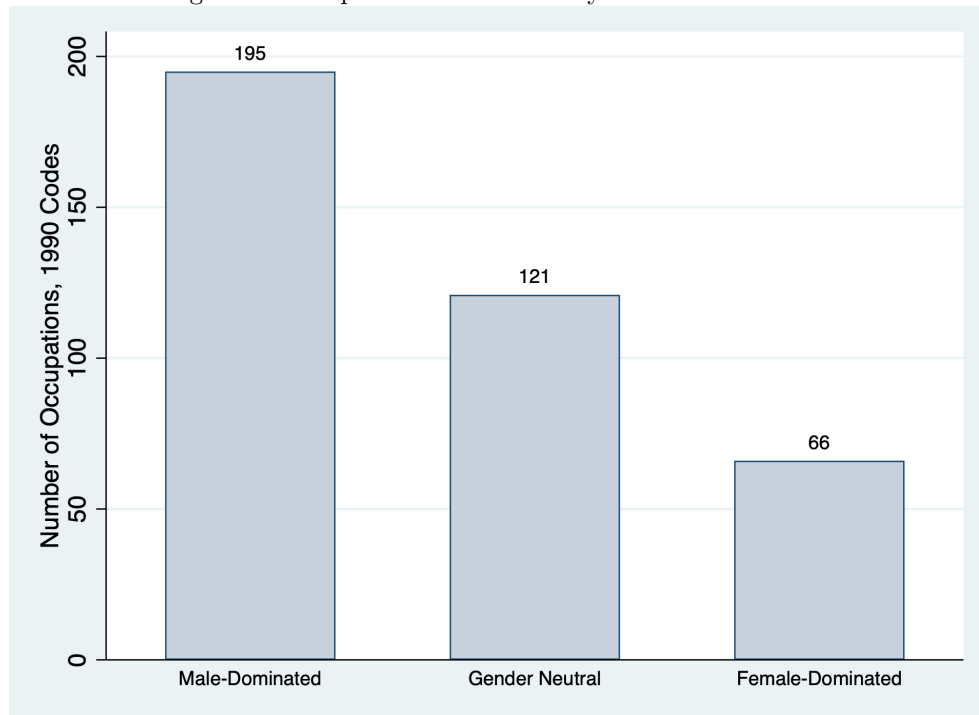
they constitute a significant portion of the workforce. In using the 70% cut-off, I am able to discriminate between Duncan Indices for the market and those for just pink-collar work. Duncan Indices are constructed among pink-collar jobs by only including in the calculus occupations that were 70% or more female pre-shock. Further, I measure the average share of men in pink-collar occupations to understand the effects of trade shocks on occupational sex composition. As pink-collar jobs are less ubiquitous than other occupations, fewer commuting zones are represented in analyses of pink-collar work.

I use US-China import penetration to measure the effect of increased manufacturing competition from China. Import penetration data is sourced directly from Autor et al. [2017] using UN Comtrade and County Business Patterns data. Import penetration is a measure of industry imports as a percentage of total industry demand, and thus can highlight changes in local labor markets due to trade shocks. Import penetration is calculated by the below formula:

$$IP_{i\tau} = \frac{M_{i\tau}}{Y_{i\tau} - M_{i\tau} + X_{i\tau}}$$

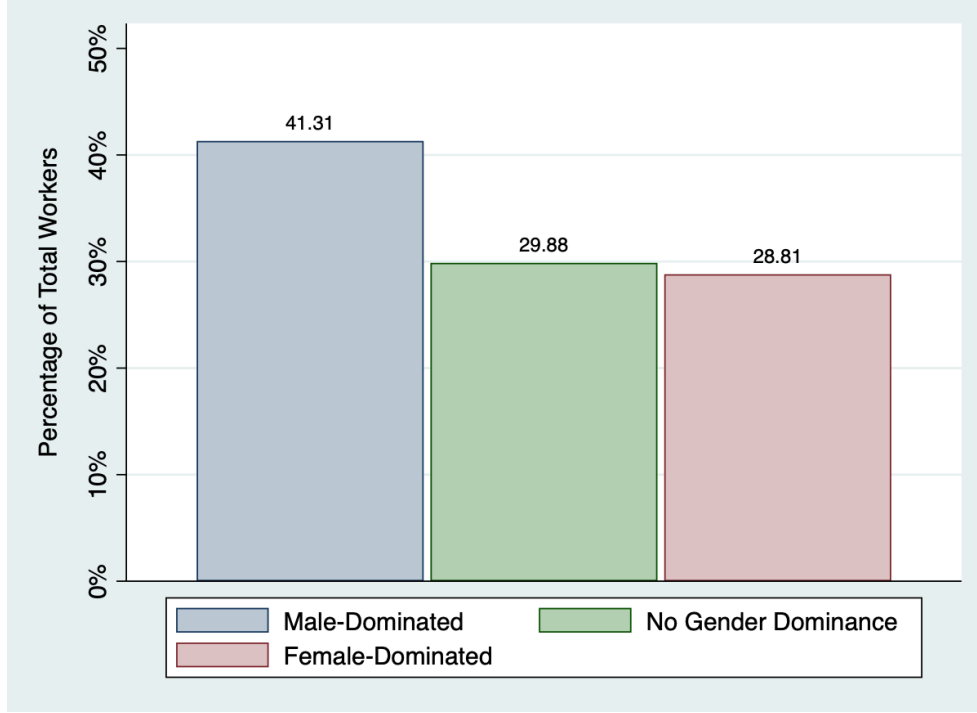
Where import penetration $IP_{i\tau}$ in commuting zone i during period τ equals imports $M_{i\tau}$ divided by GDP $Y_{i\tau}$ plus net exports $(-M_{i\tau} + X_{i\tau})$. I weight import penetration by labor shares across a commuting

Figure 4: Occupation Distribution by Gender Dominance



*Note: Consistent with all gender-dominance cut-offs in this paper, these numbers come from the 1980 Census sample, though occupational codes are harmonized to the occ1990 codes. Gender dominance is again defined by a 70% cut-off in the share of men (women) in an occupation. My analysis omits occupational codes for unknown and unemployed, as I focus specifically on occupational segregation. The “catch-all” effects of including the code for unknown occupations or unemployed persons may warp analysis of segregation since these categories do not refer to a specific occupation. Source: 1980 Census 5% state sample. [Ruggles et al., 2020].

Figure 5: Percentage of the Workforce by Occupational Gender Dominance, 1980 Census Sample



Source: 1980 Census 5% state sample. [Ruggles et al., 2020].

Note: Worker numbers based on 1980 Census sample. Percentages are based on the number of workers in each occupation, which is noted as female- or male-dominated.

zone's industries to capture the labor market effects of these shocks. My data captures the average change in import penetration in the US from Chinese trade across US commuting zones.

$$\Delta IP_{i\tau}^{cu} = \sum_j \Delta IP_{j\tau}^{cu} \times \frac{L_{ij90}}{L_{i90}}$$

Where import penetration $\Delta IP_{i\tau}^{cu}$ is the decadal change in Chinese import penetration in a commuting zone's industries, weighted by each commuting zone's initial employment in industry j in base year 1990 ($\frac{L_{ij90}}{L_{i90}}$). The variable reflects changes in import penetration over the two periods of analysis τ , 1990-2000 and 2000-2014.

Table 8 in the Appendix displays the controls I use in my regressions. Each control is commuting-zone-wide over the two periods, 1990-2000 and 2000-2014.

3.2 Summary Statistics

Population-weighted summary statistics for each of my variables of interest can be seen in Table 1.

The unit of observation for these summary statistics is one commuting zone. Consistent with previously mentioned trends, this table shows, on average, that import penetration has been rising and occupational

Table 1: Summary Statistics 1990-2014, Weighted by Commuting Zone Population Share
1990-2000

Variable by Commuting Zone	Observations	Mean	Std. Dev	Min	Max
Δ US Import Penetration	722	0.950	0.611	-0.0858	9.983
Δ Control Import Penetration	722	0.707	0.481	-0.557	6.106
Δ Occupational Segregation	722	-0.169	14.250	-100	100
Δ Occupational Segregation, Pink-Collar	415	1.355	7.253	-56.892	100
Δ Average Share of Males, Pink-Collar	415	4.344	6.138	-100	100
2000-2014					
Variable by Commuting Zone	Observations	Mean	Std. Dev	Min	Max
Δ US Import Penetration	722	1.150	0.767	-0.786	6.409
Δ Control Import Penetration	722	1.412	0.753	-0.160	6.631
Δ Occupational Segregation	722	-0.543	15.627	-100	100
Δ Occupational Segregation, Pink-Collar	415	-18.966	7.218	-80.134	58.249
Δ Average Share of Males, Pink-Collar	415	1.405	6.251	-100	100

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. *See* Ruggles et al. [2020], Autor et al. [2017].

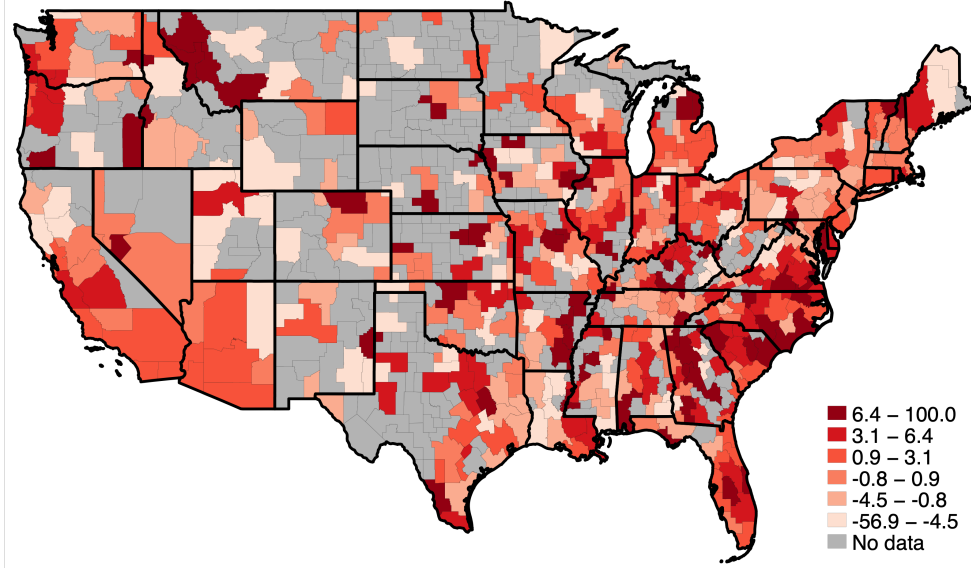
segregation has been falling, though this trend has slowed over time. Notably, a one-unit rise in import penetration is approximately equal to the average increase in import penetration over a decade. This observation is especially important within the context of my results.

Large variation in occupational segregation and share of males in pink-collar occupations comes from small commuting zones, where these measures are highly sensitive to changes in workforce composition. For example, if commuting zone 1 has a sample size of 1 and that person is a female doctor, its Duncan Index would be 100. If the next sample year, a male doctor is also sampled, the Duncan Index would be 0. These large changes among small commuting zones are down-weighted when I use population weighting, but can be seen in the extreme range in minimums and maximums for measures of gender segregation.⁵ A smaller standard deviation among segregation in pink-collar work is to be expected, as isolating pink-collar work would imply isolating higher Duncan Indices (since we expect gender segregation across pink-collar work to be skewed toward women).

Loss of commuting zones when examining occupational segregation in pink-collar jobs stems from small commuting zones where not enough pink-collar jobs existed to compute Duncan Indices across the two periods. While these missing commuting zones are consistent in both periods and accounted for in my regressions, this is surely a limitation of any analysis on pink-collar jobs only. Still, as I show in Figure 6, these commuting zones still cover all major metropolitan areas in the United States.

⁵This is further demonstrated in Figure 7 in the Appendix, which shows that most commuting zones' segregation levels are

Figure 6: Map of Change in Pink-Collar Segregation across US Commuting Zones, 1990-2000



Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020].

4 Empirical Strategy

I identify the causal effect of manufacturing import penetration from China on gender occupational segregation in local labor markets. I use both OLS and 2SLS estimation strategies to understand market-wide changes, before examining heterogeneity in these results across pink-collar jobs and regional differences. In all specifications, variables of interest were converted from decadal data to differences data across two time periods: 1990-2000 and 2000-2014 [Autor et al., 2017]. Time frames were chosen due to import penetration data availability. In using differences data across US commuting zones, I difference out fixed effects from each commuting zone that would exist in levels data.

The primary model I examine in this paper is as follows:

$$\Delta S_{i\tau}^{M,P} = \alpha_{\tau} + \beta \times \Delta IP_{i\tau}^{cu} + X'_{i\tau} + \epsilon_{i\tau} \quad (1)$$

Where $\Delta S_{i\tau}$ represents decadal change in gender occupational segregation ΔS both market-wide (M) and across pink-collar jobs (P) within commuting zone i in period τ . In this equation, α_{τ} represents time effects during period τ . $\Delta IP_{i\tau}$ represents decadal change in manufacturing import penetration across the commuting zone i in period τ . The control vector $X'_{i\tau}$ represents start-of-period covariates within commuting zones, including racial controls, share of population that is foreign-born, the employment rate of working-age women, controls for automation and offshoring in occupations, and share of the population with a college

around 50-60%, but extreme values are more common in less populous areas.

education.⁶ In all specifications, results are weighted by start-of-period commuting zone population. This is especially important in my analysis, as segregation indices in small commuting zones can be extreme.

Duncan Indices measure overall market segregation, but do not identify whether shifts operate through changes in the sex composition of occupations or changes in the occupation mix of a commuting zone. To examine whether changes in segregation associated with trade shocks are related to male entries to pink-collar work, I use a similar model, shown below:

$$\Delta SOM_{i\tau}^P = \alpha_\tau + \beta \times \Delta IP_{i\tau}^{cu} + X'_{i\tau} + \epsilon_{i\tau} \quad (2)$$

This model is nearly identical to Equation 1, except the dependent variable $\Delta SOM_{i\tau}^P$ represents decadal change in the average share of men (ΔSOM) employed in pink-collar occupations (P) within commuting zone i in period τ . By looking at the share of men in pink-collar jobs, I am able to better estimate whether trade shocks affect male entries to pink-collar jobs, or if changes in the Duncan Index are caused by other market phenomena.

Following the approach of Autor et al. [2017], I use an instrumental variable approach to identify the effect of Chinese trade competition on gender segregation. Import penetration can capture both exogenous changes due to internal growth in Chinese economy and endogenous changes in demand in US markets. For example, if a demand shock occurs in pink-collar work, the observable result would not only be increased import growth, but also rising wage and employment for people in those occupations. This change would thus be biased toward women, and may result in a biased OLS estimation. To avoid this, I use an instrumental variable approach to rule out potential bias that could arise in a basic OLS model.

I instrument Chinese import penetration of other highly developed countries $\Delta IP_{i\tau}^{coth}$ on China-US import penetration $\Delta IP_{i\tau}^{cu}$, to isolate the exogenous component of manufacturing trade shocks, following the approach of Autor et al. [2017].⁷

$$\Delta IP_{i\tau}^{coth} = \sum_j \Delta IP_{j\tau}^{coth} \times \frac{L_{ij80}}{L_{i80}}$$

$$\Delta IP_{i\tau}^{cu} = \alpha_\tau + \beta \times \Delta IP_{i\tau}^{coth} + X'_{i\tau} + \epsilon_{i\tau}$$

Unlike US-China import penetration ($\Delta IP_{i\tau}^{cu}$), this instrument ($\Delta IP_{i\tau}^{coth}$) is weighted by 1980 labor

⁶Notably, Cortes and Pan (2019) critique Autor's method of calculating routinization, as it does not entirely account for how technological change affects what makes a job susceptible to routinization.

⁷These countries include Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland.

shares to avoid simultaneity bias.

5 Results

5.1 The Effect of Trade Shocks on Overall Market Gender Segregation

I first examine overall effects of trade shocks on segregation, before examining heterogeneity in results, using OLS and 2SLS models described in Equation 1. Table 2 shows the results of this regression.

Table 2: Impact of Chinese Trade Shock on Market-Wide Occupational Gender Segregation across US Commuting Zones, 1990-2014.

	(1)	(2)	(3)	(4)
	$\Delta Segregation$ Market-Wide OLS	$\Delta Segregation$ Market-Wide OLS	$\Delta Segregation$ Market-Wide 2SLS	$\Delta Segregation$ Market-Wide 2SLS
$\Delta USImportPenetration$	0.068 (0.521)	-0.247 (0.558)	0.057 (0.709)	-0.257 (0.685)
constant	-0.460 (0.675)	0.362 (9.202)	-0.448 (0.849)	0.366 (9.040)
Controls?	no	yes	no	yes
Observations	1444	1444	1444	1444
Adjusted R^2	-0.001	0.022	-0.001	0.022

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations = 1444, which represents the 722 commuting zones over 2 periods ($722 \times 2 = 1444$). Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state.

In Models 1 and 2, I estimate the impact of Chinese trade shocks on segregation using OLS estimates, whereas in Models 3 and 4 I use 2SLS estimates. OLS estimates are shown as a point of reference, though, in this specification, these values are not statistically different from 2SLS estimates.

In Models 1 and 3, a one-unit rise in import penetration (or the average decadal change in import penetration) is associated with a slight rise in gender segregation, whereas, when controls are added in Models 2 and 4, an increase of the average decadal change in import penetration import penetration is associated with a negative pressure on overall market segregation. However, in all four of these models, the magnitudes of the coefficients on import penetration are incredibly small. Even in model 4, these results would imply that a one-decade increase in import penetration would only make it such that 0.257 percentage points fewer workers would need to change jobs for there to be perfect gender integration. Therefore, with such small coefficients, these results suggest that increased import penetration is not associated with a significant change in gender segregation. Further, the standard errors on these values are larger than the

coefficients themselves, suggesting that, even if these effects had a greater magnitude, they would not be significantly different than zero.

While these results show that overall market segregation may not be significantly influenced by growing Chinese import penetration, they do not suggest that import penetration has no significant effect whatsoever on occupational segregation. To evaluate heterogeneity in these results that may arise due to different gender compositions in a commuting zone’s manufacturing industries, I followed the approach of Autor et al. [2017] by examining the gender-specific components of trade shocks. I do so by using an interaction term with initial male or female employment share in each commuting zone’s industries and import penetration. These measures of import penetration $\Delta IP_{i\tau}^{cu,m}$ and $\Delta IP_{i\tau}^{cu,f}$ isolate the gender-specific effects of import penetration.

In Table 3, I examine 2SLS models focusing on these gender-specific effects.

Table 3: Gender-Specific Impact of Chinese Trade Shock on Market-Wide Occupational Gender Segregation across US Commuting Zones, 1990-2014

	(1) $\Delta Segregation$ Market-Wide 2SLS	(2) $\Delta Segregation$ Market-Wide 2SLS
$\Delta US Import Penetration$	-0.257 (0.685)	
$\Delta US Import Penetration \times \text{Male Emp. Share}$		-7.826*** (2.188)
$\Delta US Import Penetration \times \text{Female Emp. Share}$		8.525*** (2.579)
constant	0.366 (9.040)	-2.813 (10.266)
Controls?	yes	yes
Observations	1444	1444
Adjusted R^2	0.022	0.012

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. Standard errors are clustered by commuting zone’s state.

When this overall effect is separated by its gender specific shocks, the results of this model change substantially. In Model 1, I show the overall effect of import penetration on occupational segregation. However, in Model 2, I examine the gender-specific elements of trade shocks. Model 2 shows that the gender-specific effects of import penetration on occupational segregation are significant, and may be obscured in the overall market image we are presented with in Model 1.

Model 2 shows that a one unit increase in male-specific import penetration is associated with a signifi-

cant negative pull on gender segregation ($\beta = -7.826$) whereas a similar increase in female-specific import penetration seems to exert a positive influence on gender segregation ($\beta = 8.525$). Both of these coefficients are significant at the 1% level. These estimates imply that trade shocks have varied effects based on gender: gender-specific trade shocks seem to increase segregation when isolating female-specific shocks and decrease segregation when examining male-specific shocks.

Occupational segregation can be decomposed into the occupation mix and sex composition effects [Blau et al., 2013]. The first of these implies that the occupations represented in the Duncan Index have changed. The sex composition effect, on the other hand, implies that the actual number of men and women in certain jobs has changed. Applying this to my results, this regression can be explained in two ways. First, these effects may be caused by a change in the occupation mix of an economy. As blue-collar work declines, male-dominated jobs are no longer accounted for in Duncan Indices, which makes it appear that segregation is falling. Likewise, female-dominated occupations are rapidly growing, so these jobs skew the Duncan Index toward a higher level of segregation. Second, these effects may be attributable to sex composition changes within occupations. The negative coefficient on male-specific shocks may suggest men are moving toward female-dominated industries as traditionally-male jobs decline, and the positive coefficient on female-specific shocks may similarly imply that more women are moving toward pink-collar work.

To further investigate whether trade shocks affect the sex composition or occupation mix of an economy, I isolate segregation across pink-collar jobs.

5.2 Examining Differences across Industries: Zooming in on Pink-Collar Work

Using the same model as the previous section, I regress pink-collar gender segregation on changes in import penetration from 1990-2014 to understand how trade specifically affects pink-collar jobs. We expect that pink-collar jobs are highly segregated, as they are, by definition, 70% or more female according to the 1980 Census. I use a 2SLS estimation to understand these effects in Table 4:

In Table 4, Model 1 is identical in strategy to previous analyses of overall market segregation and import penetration, but only observes commuting zones where data on pink-collar segregation exists. Examining the coefficient on import penetration in this model, we again do not see a significant relationship between overall occupational segregation and import penetration. Even if the coefficient on import penetration were significant, its magnitude is incredibly small.

However, when examining overall import penetration and segregation in pink-collar work in Model 2, we see that a one unit increase in import penetration is associated with a significantly negative effect on occupational segregation in pink-collar jobs. This suggests that, when we look specifically at pink-collar

Table 4: Impact of Manufacturing Trade Shock on Occupational Gender Segregation in Pink-Collar Jobs across US Commuting Zones, 1990-2014

	(1) $\Delta Segregation$ Market-Wide 2SLS	(2) $\Delta Segregation$ Pink-Collar 2SLS	(3) $\Delta Segregation$ Pink-Collar 2SLS
$\Delta USImportPenetration$	0.170 (0.483)	-9.205*** (1.439)	
$\Delta USImportPenetration \times \text{Male Emp. Share}$			-28.912*** (7.473)
$\Delta USImportPenetration \times \text{Female Emp. Share}$			15.328** (6.896)
constant	-4.265 (4.585)	83.025*** (14.925)	78.340*** (14.826)
Controls?	yes	yes	yes
Observations	830	830	830
Adjusted R^2	0.006	0.244	0.157

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations = 830, which represents the 415 commuting zones over 2 periods ($415 \times 2 = 830$). Fewer commuting zones are shown in this analysis compared to previous analyses due to data availability. Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state.

work, Chinese trade competition may contribute to a decline in occupational segregation in female-dominated sectors.

These results are even more pronounced in Model 3, where I isolate the gender-effects of import penetration on pink-collar segregation. In this model, male-specific trade shocks are associated with a 28.912 point drop in pink-collar occupational segregation. An effect of this magnitude is especially significant, as occupational segregation only varies between zero and 100. Model 3 also suggests that, when isolating female-specific components of the trade shock, these changes are associated with a significantly large increase in segregation.

The results from Table 4 show that male-specific trade shocks are associated with declining pink-collar occupational segregation, but these results alone do not explain where changes in segregation come from. Since there is no significant evidence that gender segregation in pink-collar work is changing due to male entries, the results of Table 4 may imply that changes in occupational segregation are not coming from China-shock-driven male entries to pink-collar work, but from changes in commuting zones' occupational mix as blue-collar work declines and pink-collar work booms. To investigate whether these results can be explained by changes in the gender composition of pink-collar jobs, I look into the effect of trade shocks on the average share of males in pink-collar jobs below.

5.3 Disentangling the Sex Composition and Occupation Mix Effects

As noted above, simply examining changes in occupational segregation cannot speak to whether these changes are due to men entering pink-collar work. To identify the effect of trade shocks on the sex composition of pink-collar jobs, I look specifically at the share of men within female-dominated occupations. I show these results in Table 5.

Table 5: Impact of Manufacturing Trade Shock on Average Share of Males in Pink-Collar Occupations across US Commuting Zones, 1990-2014.

	(1) ΔSOM Pink-Collar 2SLS	(2) ΔSOM Pink-Collar 2SLS
$\Delta USImportPenetration$	-1.213*** (0.379)	
$\Delta USImportPenetration \times \text{Male Emp. Share}$		-3.543** (1.585)
$\Delta USImportPenetration \times \text{Female Emp. Share}$		1.687 (2.020)
constant	13.728*** (3.826)	13.174*** (4.151)
Controls?	yes	yes
Observations	830	830
Adjusted R^2	0.025	0.021

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations = 830, which represents the 415 commuting zones over 2 periods ($415 \times 2 = 830$). Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state.

From these results, increased import penetration contributes to a decline in the share of men in pink-collar jobs, rather than driving men to these occupations. In Model 1, a one-unit rise in import penetration is associated with a 1.213 point drop in the average share of men in pink-collar jobs. This value implies that increased trade competition from China is associated with men composing less of the pink-collar workforce. In Model 2, where I isolate the gender-specific components of these trade shocks, this result is even more pronounced. Male-specific trade shocks are associated with a 3.543 point drop in the average share of men across pink-collar occupations in a commuting zone. While the magnitude of both of these results is relatively small, the fact that the relationship is negative is significant.

These results suggest that, if anything, trade shocks may decrease the share of men in pink-collar jobs. There are several reasons this may be occurring. First, men may resist entering pink-collar jobs, even as opportunities in blue-collar work decline. Pink-collar work may represent an identity mismatch, or still may

be too financially- or socially-undervalued compared to blue-collar jobs. On the other hand, women may still dominate these fields because, as service and clerical sectors grow, they retain a comparative advantage in occupational tasks [Ngai and Petrongolo, 2017].

Nonetheless, increased trade competition from China is associated with a decline in the average share of men in pink-collar jobs, even though it is likewise associated with a fall in gender occupational segregation. Combining these findings, my results suggest that declining pink-collar occupational segregation as a result of increased trade competition is driven by a change in the occupation mix in local labor markets, rather than male entries to pink-collar jobs.

5.4 Examining Regional Variation

To further investigate how my results may vary across the US, I examine changes across Census regional divisions to see if certain regions, states, and commuting zones' occupational segregation levels are particularly impacted by shocks. Following the same specification as in 5.1, I interact regional dummy variables with import penetration to examine heterogeneity across Census divisions in Table 6.

Model 1 represents segregation across all markets, Model 2 represents segregation across all markets for which enough data is available to calculate pink-collar occupational segregation, and Model 3 shows segregation in pink-collar jobs. Models 1 and 3 are the primary estimations of interest, though Model 2 allows for comparison between pink-collar occupational segregation and overall occupational segregation effects.

In Model 1, similar to previous results, the impact of trade shocks on occupational segregation is not significantly different from zero. The only region in which this effect is significant is the West-North Central division, which shows that import penetration is associated with a slightly positive pull on occupational segregation. Since it is unlikely that occupations in this region are becoming more segregated within occupations, this is likely due to growth of more segregated occupations.

However, when identifying the effect of trade shocks on pink-collar occupational segregation, we see a statistically significant negative relationship in all regions except the Mid Atlantic, South Atlantic, Mountain, and Pacific regions. In all other regions, a one unit increase of import penetration is associated with a significant decline in pink-collar occupational segregation. These effects are most pronounced in the East-North Central and West-North Central divisions, where import penetration is associated with a significantly large drop in occupational segregation. These results are consistent with the fact that these regions have been hit especially hard by trade shocks in manufacturing. Still, as we see in analyses above, this is not necessarily attributable to male entries in pink-collar work.

Table 6: Regional Differences in the Impact of Manufacturing Trade Shock on Occupational Gender Segregation across US Commuting Zones, 1990-2014

	(1)	(2)	(3)
	$\Delta Segregation$ Market-Wide 2SLS	$\Delta Segregation$ Market-Wide 2SLS	$\Delta Segregation$ Pink-Collar 2SLS
$\Delta USImportPenetration$	-1.205 (0.967)	0.188 (0.658)	-6.979*** (1.545)
$\Delta USImportPenetration \times \text{Mid. Atl.}$	0.379 (0.637)	-0.152 (0.491)	-4.104 (3.418)
$\Delta USImportPenetration \times \text{E. N. Cen.}$	0.979 (0.778)	0.049 (0.523)	-7.981*** (2.536)
$\Delta USImportPenetration \times \text{W. N. Cen.}$	2.810** (1.249)	1.820 (1.593)	-4.196** (1.722)
$\Delta USImportPenetration \times \text{S. Atl.}$	0.682 (0.862)	-0.178 (0.801)	-2.141 (1.979)
$\Delta USImportPenetration \times \text{E. S. Cen.}$	2.112 (1.308)	0.403 (1.296)	-3.646** (1.741)
$\Delta USImportPenetration \times \text{W. S. Cen.}$	-0.049 (0.764)	-0.444 (0.625)	-3.550** (1.580)
$\Delta USImportPenetration \times \text{Mountain}$	0.369 (1.254)	0.291 (1.115)	-2.266 (1.558)
$\Delta USImportPenetration \times \text{Pacific}$	0.527 (0.866)	-0.200 (0.627)	2.253 (1.870)
constant	1.358 (10.124)	-3.005 (4.980)	63.742*** (13.273)
Controls?	yes	yes	yes
Observations	1444	830	830
Adjusted R^2	0.018	-0.002	0.231

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations for market-wide = 1444 (pink-collar = 830), which represents the 722 (415) commuting zones over 2 periods. Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state. Omitted region is Northeast.

To examine regional variation in trade-shock-driven male entries to pink-collar occupations, I regress the average share of males in pink-collar jobs on these regional interaction terms. These results are shown in Table 7.

Model 2 of Table 7 shows that the regional effect of import penetration on the average share of males in pink-collar work is small in magnitude. Across these measures, in the three regions whose coefficients on import penetration are significant—the Northeast, East-North Central, and West-North Central divisions—the sign on the coefficient is negative. In these three regions, this effect is largest in the East-North Central region. This again suggests that, even as trade shocks are associated with a decline in occupational segregation in pink-collar work, this shift is not clearly caused by trade-driven male entries to pink-collar jobs. While

Table 7: Regional Differences in the Impact of Manufacturing Trade Shock on Average Share of Males in Pink-Collar Work across US Commuting Zones, 1990-2014

	(1) ΔSOM Pink-Collar 2SLS	(2) ΔSOM Pink-Collar 2SLS
$\Delta USImportPenetration$	-1.213*** (0.379)	-1.182** (0.460)
$\Delta USImportPenetration \times \text{Mid. Atl.}$		-0.745 (0.561)
$\Delta USImportPenetration \times \text{E. N. Cen.}$		-0.889*** (0.344)
$\Delta USImportPenetration \times \text{W. N. Cen.}$		-0.875** (0.413)
$\Delta USImportPenetration \times \text{S. Atl.}$		0.294 (0.398)
$\Delta USImportPenetration \times \text{E. S. Cen.}$		-0.003 (0.621)
$\Delta USImportPenetration \times \text{W. S. Cen.}$		-0.178 (0.572)
$\Delta USImportPenetration \times \text{Mountain}$		0.201 (0.447)
$\Delta USImportPenetration \times \text{Pacific}$		0.522 (0.490)
constant	13.728*** (3.826)	10.544** (4.542)
Controls?	yes	yes
Observations	830	830
Adjusted R^2	0.025	0.019

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations = 830, which represents 415 commuting zones over 2 periods ($415 \times 2 = 830$). Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state. Omitted region is Northeast.

increased US-China import penetration seems to contribute to a decline in pink-collar gender segregation, my results suggest this is not due to male entries to pink-collar occupations. The evidence indicates trade-shock-driven changes in occupational segregation are driven by the decline of highly-segregated occupations, such as male-dominated manufacturing jobs.

6 Conclusion

Starting in the 1990s, China's shift to an open-market economy rapidly affected high-income nations, while at the same time changes in gender segregation began to slow. In this paper, I examined the effects of Chinese

trade shocks in US local labor markets to explain changes in gender occupational segregation, particularly in pink-collar jobs. I test whether trade shocks induce male entry into female-dominated occupations, as a large loss in blue-collar occupations would eliminate job prospects for men. Faced with unemployment, I hypothesized that men may move to pink-collar jobs. I find that, while there is no overall effect on gender occupational segregation associated with Chinese trade shocks, when isolating the gender specific components of these shocks, male-specific trade shocks are associated with a fall in overall market gender occupational segregation.

When focusing on segregation across pink-collar jobs, these effects are even more pronounced. Rising import penetration is associated with a significant decline in pink-collar occupational segregation, and the magnitude of this effect is even larger when isolating the male-specific components of this shock. However, this result does not necessarily suggest occupational segregation is falling due to male entries to pink-collar jobs. Since occupational segregation can change due to occupational composition and gendered movement in the job market, I seek to disentangle these effects by looking at the average share of men in a commuting zone's pink-collar occupations. The results of this analysis suggest that, if anything, import penetration is marginally decreasing the average share of males in pink-collar work. This result suggests that declining segregation is more likely due to changes in a commuting zone's occupation mix rather than various occupations' sex composition.

Literature on gender attachment to occupational identity suggests that men would resist movement to pink-collar jobs, since these jobs are socially below male-jobs, financially undervalued, and incompatible with their own gender identity. Despite incentives of growth in pink-collar jobs and manufacturing decline, it is not clear that men are entering pink-collar jobs due to increased Chinese trade competition. Rather, trade specific changes in gender segregation may be caused by a change in the occupational mix in commuting zones, as blue-collar jobs decline and pink-collar jobs grow.

Decreasing gender occupational segregation should be a priority in closing the economic gap between men and women, as it has numerous effects both on earning potential and social identity. While the economic effects of Chinese trade shocks have greatly influenced the landscape of blue-collar work in the United States, it is not clear that these effects are similarly working to change pink-collar jobs. Further research, such as examining the effects of trade shocks on individual pink-collar jobs or examining import penetration in other sectors, remains important to investigating these trends.

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Appendix

Additional Descriptives

Table 8: Summary Statistics for Covariates
1990-2000

Variable by Commuting Zone	Observations	Mean	Std. Dev	Min	Max
Labor Share of Population, Black	722	12.115	9.884	0	60.911
Labor Share of Population, Asian	722	2.659	3.147	0	15.617
Labor Share of Population, Hispanic	722	9.006	12.149	0	93.529
Labor Share of Population, Other Race	722	4.713	5.989	0	75.374
Labor Share of Population, College-Educated	722	48.021	8.113	19.944	69.650
Labor Share of Population, Routine Tasks	722	32.228	2.834	19.992	37.748
Labor Share of Population, Offshored Tasks	722	0.112	0.462	-1.491	1.025
Labor Share of Population, Foreign-Born	722	10.146	10.272	0.385	39.944
Female Employment Rate, Working-Age	722	63.624	5.535	33.243	77.299
2000-2014					
Variable by Commuting Zone	Observations	Mean	Std. Dev	Min	Max
Labor Share of Population, Black	722	12.378	10.136	0	66.145
Labor Share of Population, Asian	722	3.475	3.766	0	19.405
Labor Share of Population, Hispanic	722	12.585	14.169	0.186	95.012
Labor Share of Population, Other Race	722	8.772	7.900	0.254	81.200
Labor Share of Population, College-Educated	722	53.453	7.479	26.320	70.555
Labor Share of Population, Routine Tasks	722	31.872	2.392	22.227	36.656
Labor Share of Population, Offshored Tasks	722	-0.0131	0.512	-1.636	1.240
Labor Share of Population, Foreign-Born	722	14.674	12.814	0.621	48.908
Female Employment Rate, Working-Age	722	64.334	5.541	41.339	79.606

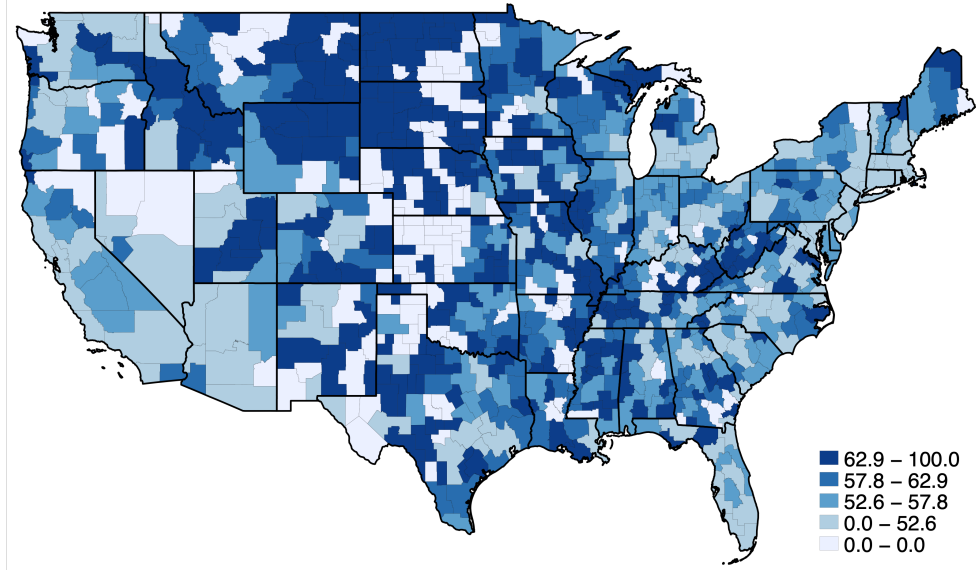
Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples

Table 9: Female-Dominated Occupations in 1980, Sorted by Percentage Female

N	Occupational Title, 1990 Codes	Female Share	N	Occupational Title, 1990 Codes	Female Share
1	Secretaries	98.8%	34	Librarians	83.9%
2	Dental hygienists	98.8%	35	Payroll and timekeeping clerks	83.5%
3	Dental assistants	97.9%	36	Eligibility clerks for government programs; social welfare	83.2%
4	Licensed practical nurses	96.9%	37	General Office Clerks	83.1%
5	Typists	96.9%	38	Telephone operators	83.0%
6	Kindergarten and earlier school teachers	96.7%	39	Food counter and fountain workers	82.7%
7	Receptionists	96.4%	40	Proofreaders	82.1%
8	Registered nurses	96.2%	41	File clerks	82.0%
9	Private household cleaners and servants	95.0%	42	Library assistants	81.1%
10	Textile sewing machine operators	95.0%	43	Solderers	80.9%
11	Dressmakers and Seamstresses	94.3%	44	Housekeepers, maids, butlers, stewards, and lodging quarters cleaners	80.7%
12	Child Care Workers	94.1%	45	Information clerks, n.e.c.	80.6%
13	Home Economics Instructors	93.2%	46	Interviewers, enumerators, and surveyors	80.3%
14	Data Entry Keyers	92.9%	47	Kitchen workers	79.8%
15	Occupational therapists	92.8%	48	Primary school teachers	78.4%
16	Teacher's aides	92.4%	49	Shoemaking machine operators	78.3%
17	Stenographers	91.9%	50	Public transportation attendants and inspectors	78.1%
18	Health record tech specialists	91.7%	51	Pressing machine operators (clothing)	78.0%
19	Bank tellers	91.3%	52	Dancers	77.8%
20	Bookkeepers and accounting and auditing clerks	90.4%	53	Winding and twisting textile/apparel operatives	77.0%
21	Dietitians and nutritionists	90.3%	54	Retail sales clerks	76.6%
22	Waiter/waitress	90.1%	55	Statistical clerks	75.8%
23	Speech therapists	89.8%	56	Clinical laboratory technologtes and technicians	75.7%
24	Hairdressers and cosmetologists	89.6%	57	Personal service occupations, n.e.c.	75.1%
25	Welfare service aides	89.3%	58	Physical therapists	74.3%
26	Nursing aides, orderlies, and attendants	89.1%	59	Radiologic tech specialists	73.4%
27	Billing clerks and related financial records processing	88.9%	60	Adjusters and calibrators	72.1%
28	Human resources clerks, except payroll and timekeeping	88.1%	61	Office machine operators, n.e.c.	72.0%
29	Sales demonstrators / promoters / models	87.4%	62	Crossing guards and bridge tenders	71.2%
30	Cashiers	85.6%	63	Special education teachers	70.5%
31	Records clerks	85.4%	64	Therapists, n.e.c.	70.4%
32	Health aides, except nursing	85.0%	65	Packers and packagers by hand	70.4%
33	Graders and sorters of agricultural products	84.6%	66	Correspondence and order clerks	70.0%

Source: 1980 Census 5% state samples. See Ruggles et al. [2020].

Figure 7: Map of Levels of Gender Segregation across US Commuting Zones, 2014



Source: 2013-15 ACS samples. See Ruggles et al. [2020].

Robustness: Results Including Unemployed Individuals and Individuals Not in the Labor Force

In the below tables, I replicate my results from Tables 2 - 7, but include in my analysis individuals of all employment status in Census samples. These results, though not leading to significantly different conclusions than those posited above, may reflect nuances in the labor force not captured in previous analyses. As many economists have noted, the China shock has seemed to have a significant effect on male unemployment in manufacturing [Autor et al., 2017]. The below analysis measures segregation across all sampled men and women, which may show differences in segregation driven by individuals' occupational identity, regardless of their employment status.

Table 10: Impact of Manufacturing Trade Shock on Market-Wide Occupational Gender Segregation, Including Unemployed Individuals and Individuals Not in the Labor Force, across US Commuting Zones, 1990-2014

	(1)	(2)	(3)	(4)
	$\Delta Segregation$	$\Delta Segregation$	$\Delta Segregation$	$\Delta Segregation$
	Market-Wide	Market-Wide	Market-Wide	Market-Wide
	OLS	OLS	2SLS	2SLS
$\Delta USImportPenetration$	0.953	-0.247	0.751	0.338
	(0.707)	(0.558)	(0.887)	(0.797)
constant	-1.436*	0.362	-1.219	-8.020
	(0.734)	(9.202)	(0.940)	(9.685)
Controls?	no	yes	no	yes
Observations	1444	1444	1444	1444
Adjusted R^2	0.001	0.022	0.001	0.026

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations = 1444, which represents the 722 commuting zones over 2 periods ($722 \times 2 = 1444$). Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state.

Table 11: Gender-Specific Impact of Manufacturing Trade Shock on Market-Wide Occupational Gender Segregation, Including Unemployed Individuals and Individuals Not in the Labor Force, across US Commuting Zones, 1990-2014

	(1)	(2)
	$\Delta Segregation$	$\Delta Segregation$
	Market-Wide	Market-Wide
	2SLS	2SLS
$\Delta USImportPenetration$	0.338	
	(0.797)	
$\Delta USImportPenetration \times \text{Male Emp. Share}$		-3.852*
		(2.325)
$\Delta USImportPenetration \times \text{Female Emp. Share}$		5.199*
		(2.997)
constant	-8.020	-9.780
	(9.685)	(10.094)
Controls?	yes	yes
Observations	1444	1444
Adjusted R^2	0.026	0.031

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations = 1444, which represents the 722 represented commuting zones over 2 periods ($722 \times 2 = 1444$). Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state.

Table 12: Impact of Manufacturing Trade Shock on Occupational Gender Segregation in Pink-Collar Jobs, Including Unemployed Individuals and Individuals Not in the Labor Force, across US Commuting Zones, 1990-2014

	(1) $\Delta Segregation$ Market-Wide 2SLS	(2) $\Delta Segregation$ Pink-Collar 2SLS	(3) $\Delta Segregation$ Pink-Collar 2SLS
$\Delta USImportPenetration$	0.610 (0.553)	-8.111*** (1.220)	
$\Delta USImportPenetration \times \text{Male Emp. Share}$			-26.212*** (6.314)
$\Delta USImportPenetration \times \text{Female Emp. Share}$			13.982** (5.434)
constant	-6.963 (6.555)	70.508*** (13.267)	64.488*** (14.020)
Controls?	yes	yes	yes
Observations	830	830	830
Adjusted R^2	0.005	0.201	0.130

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations = 830, which represents the 415 commuting zones over 2 periods ($415 \times 2 = 830$). Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state.

Table 13: Impact of Manufacturing Trade Shock on Average Share of Males in Pink-Collar Occupations, Including Unemployed Individuals and Individuals Not in the Labor Force, across US Commuting Zones, 1990-2014

	(1) ΔSOM Pink-Collar 2SLS	(2) ΔSOM Pink-Collar 2SLS
$\Delta USImportPenetration$	-1.463*** (0.278)	
$\Delta USImportPenetration \times \text{Male Emp. Share}$		-2.553** (1.213)
$\Delta USImportPenetration \times \text{Female Emp. Share}$		-0.133 (1.330)
constant	12.922*** (3.977)	12.560*** (4.206)
Controls?	yes	yes
Observations	830	830
Adjusted R^2	0.021	0.015

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations = 830, which represents the 415 commuting zones over 2 periods ($415 \times 2 = 830$). Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state.

Table 14: Regional Differences in the Impact of Manufacturing Trade Shock on Occupational Gender Segregation, Including Unemployed Individuals and Individuals Not in the Labor Force, across US Commuting Zones, 1990-2014

	(1) $\Delta Segregation$ Market-Wide 2SLS	(2) $\Delta Segregation$ Market-Wide 2SLS	(3) $\Delta Segregation$ Pink-Collar 2SLS
$\Delta US Import Penetration$	-0.583 (0.920)	-0.583 (0.920)	-6.601*** (1.410)
$\Delta US Import Penetration \times \text{Mid. Atl.}$	-0.090 (0.631)	-0.090 (0.631)	-3.402 (3.027)
$\Delta US Import Penetration \times \text{E. N. Cen.}$	0.476 (0.623)	0.476 (0.623)	-6.236*** (2.325)
$\Delta US Import Penetration \times \text{W. N. Cen.}$	3.053** (1.356)	3.053** (1.356)	-1.874 (1.208)
$\Delta US Import Penetration \times \text{S. Atl.}$	-0.181 (0.720)	-0.181 (0.720)	-1.457 (1.648)
$\Delta US Import Penetration \times \text{E. S. Cen.}$	3.423*** (0.950)	3.423*** (0.950)	-2.267 (1.499)
$\Delta US Import Penetration \times \text{W. S. Cen.}$	-0.233 (0.883)	-0.233 (0.883)	-2.597** (1.228)
$\Delta US Import Penetration \times \text{Mountain}$	1.114 (1.653)	1.114 (1.653)	-1.853 (1.328)
$\Delta US Import Penetration \times \text{Pacific}$	-0.551 (0.860)	-0.551 (0.860)	2.569 (1.661)
constant	-9.172 (10.531)	-9.172 (10.531)	54.286*** (12.339)
Controls?	yes	yes	yes
Observations	1444	1444	830
Adjusted R^2	0.029	0.029	0.187

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations = 830, which represents the 415 commuting zones over 2 periods (415 $\times$ 2=830). Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state.

Table 15: Regional Differences in the Impact of Manufacturing Trade Shock on Average Share of Males in Pink-Collar Work, Including Unemployed Individuals and Individuals Not in the Labor Force, across US Commuting Zones, 1990-2014

	(1) ΔSOM Pink-Collar 2SLS	(2) ΔSOM Pink-Collar 2SLS
$\Delta USImportPenetration$	-1.463*** (0.278)	-1.112*** (0.365)
$\Delta USImportPenetration \times \text{Mid. Atl.}$		-0.755 (0.481)
$\Delta USImportPenetration \times \text{E. N. Cen.}$		-1.154*** (0.345)
$\Delta USImportPenetration \times \text{W. N. Cen.}$		-0.752*** (0.275)
$\Delta USImportPenetration \times \text{S. Atl.}$		0.004 (0.341)
$\Delta USImportPenetration \times \text{E. S. Cen.}$		-0.640* (0.370)
$\Delta USImportPenetration \times \text{W. S. Cen.}$		-0.957*** (0.292)
$\Delta USImportPenetration \times \text{Mountain}$		-0.711** (0.358)
$\Delta USImportPenetration \times \text{Pacific}$		0.264 (0.355)
constant	12.922*** (3.977)	10.987** (4.399)
Controls?	yes	yes
Observations	830	830
Adjusted R^2	0.021	0.015

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: 1980 and 1990 Census 5% state samples, the 2000 Census 5% sample, and 2013-15 ACS samples. See Ruggles et al. [2020], Autor et al. [2017].

Notes: Parentheses under values represent standard errors. Observations = 830, which represents the 415 commuting zones over 2 periods (415×2=830). Controls include demographics, education, routine/offshorable task shares, and female employment rates, as shown in Table 8. All regressions are weighted by start-of-period commuting zone share of population. Standard errors are clustered by commuting zone's state.